

Lesson 6.8: Lab, Cheesy Pasta Marinara With Role Cards

Unit: Unit 6, From Farm to Table (Food Systems and Production) **Topic:** 6.2 Food Preparation Methods and Science **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 1 class period of 40 minutes (this is Lab 3 of the year) **Room:** FACS lab with three or four kitchen stations of about seven students (a no-kitchen version is below)

Standards

- NYS module line: NYS FSP 1. Basic Culinary Skills a) Apply basic culinary and management skills to the preparation of nutritious food
- NYS module line: NYS FSP 1. Basic Culinary Skills c) Demonstrate safety and sanitation procedures when handling food and tools
- NYS module line: NYS FSP 1. Basic Culinary Skills e) Prepare a simple recipe
- NYS module line: NYS FSP 5. Consumer Resources and Finance d) Apply time management, organizational, and process skills to prioritize tasks and achieve goals
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE theme line: CTE CIR 3. Workplace Communication f) Demonstrate effective communication skills in a group setting to accomplish a task
- NYS FACS Learning Standard 1 (Intermediate), Key Idea: Students will use an understanding of the elements of good nutrition to plan appropriate diets for themselves and others. They will know and use the appropriate tools and technologies for safe and healthy food preparation.; Performance Indicator A: Students understand the relationships among diet, health, and physical activities; evaluate their own eating patterns; and use appropriate technology and resources to make food selections and prepare simple, nutritious meals.; Performance Indicator B: Students apply principles of food safety and sanitation.
- CDOS Standard 3a Universal Foundation Skills: Managing Resources
- National FCS 3.0: 14.3.3 Show the ability to select, store, prepare, and serve food that is nourishing and looks good on the plate.
- National FCS 3.0: 8.5.7 Prepare fruits, vegetables, starches, legumes, dairy, fats, and oils safely and with professional technique.
- National FCS 3.0: 14.4.2 Examine safety and sanitation practice.

(Four module lines, one more than usual, because FSP 5. d) is the scheduling work on the critical path and it is graded here. This lesson demonstrates a second time what Unit 1 taught: FSP 1. c) and e) repeat N&W 1. b) and d) almost word for word, and the standards map lists both.)

What is new here, and what is a repeat

Unit 1 taught the four phases, the eight roles, the six-step sanitation routine, and the station check. Those are not retaught today. Students already own them, and the lesson counts on it.

What is new in this lab: boiling water is the first real critical path in the course. Water takes about eight minutes to come to a boil, pasta takes eight to ten more, and nothing else can finish until the pasta is drained. Every other task in the room has to fit around that one. That is the skill being taught, and it is the reason the flowchart matters more than the recipe.

Enduring understanding and essential question

Big idea: In a kitchen, one task sets the clock for everybody. Find that task, start it first, and build everything else around it. Miss it, and seven people stand around waiting. **Essential question:** Which single task decides whether this station eats at minute 28 or at minute 36?

Objectives

By the end of this lesson, students will be able to:

1. (Low: understand) Given the station card and their role card, state their own role's tasks in each of the four phases and name the critical path task and why it is the critical path.
2. (Mid: apply) In a team of about seven, prepare cheesy pasta with marinara: measure the salt and the cheese, boil and drain the pasta on schedule, heat the sauce without scorching it, and serve seven portions, following the six-step sanitation routine.
3. (High: evaluate) After the lab, judge whether the station kept the flowchart, name the one task that fell behind or got ahead, and recommend one change in the schedule for the next lab, naming a role and a minute.

Vocabulary

Tier 2 (general academic): schedule, prioritize, critical, evaluate, adjust Tier 3 (FACS): critical path, rolling boil, al dente, colander, drain, simmer, scorch, portion. See the unit vocabulary list (Vocabulary 06.md) for Spanish.

Materials and setup

Per station of about seven students:

- 1 lb pasta, spaghetti or elbows, store brand
- 1 jar marinara sauce, 24 oz, store brand
- 2 bags shredded mozzarella, 8 oz each (1 1/2 bags is enough; the extra covers a second station's shortfall)
- 1 Tbsp salt for the pasta water, from the room's pantry

- Tools: 1 large pot, 1 small saucepan, 1 colander, 1 large serving bowl, 1 wooden spoon, 1 large serving spoon or tongs, measuring spoons, 1 dry measuring cup, 7 plates, 7 forks, napkins, 2 oven mitts, 1 trivet, 1 sanitizing spray and 2 cloths
- Handouts: Handout 06.08 - Station Card Cheesy Pasta Marinara.md (one per station and one per student), Handout 06.08 - Role Cards Lab 3.md (one set per station), Handout 01.08 - Lab Reflection.md (the same reflection sheet used all year, one per student)
- The 40-minute flowchart in five-minute blocks on the board from the slides, a visible timer, aprons and hair ties, the clean-up zone chart taped at each station
- Rubric 01 - Lab Rubric.md on a clipboard, one group grid per class
- Slides: Slides 06.08 - Lab, Cheesy Pasta Marinara With Role Cards
- Setup notes: do not preheat anything. Students fill and carry the empty pot to the burner and turn on the heat; I carry nothing and turn on nothing, and I drain every pot. Put the grocery bags at each station so unpacking is part of prep. Fill the pot to about three quarters, not to the top, or it takes too long and it is too heavy. Have the colander already in the sink before the bell. Sanitation routine, all six steps, in this order: hair back, sleeves up, wash hands, sanitize surfaces, clean as you go, station check before dismissal. Allergy check the day before: wheat in the pasta, dairy in the cheese, and some marinara jars carry a soy or a “made on shared equipment” note. Substitutions are in Differentiation.

Pre-assessment

The Lab 2 rubric scores from Unit 1 Lesson 1.13 and the recommendations students wrote on that lab’s reflection. Read the recommendations the night before and assign roles to fix what they named. Today’s do now is the second read: a student who cannot name the critical path in the do now gets the flowchart pointed at before the timer starts.

Do now and hook (Time: 3 min, inside the prep phase)

On the board: “Hair back. Sleeves up. Wash hands. Sanitize your surface. Role card in your pocket. Then stand at your station with your hands empty. The timer starts when every station is standing.”

On the second board, in one line: “Which task in this recipe takes the longest, and what cannot start until it is done?”

Hook: the timer. Forty minutes on the board, and the flowchart beside it with the pasta bar drawn across almost the whole cook phase. “That long bar is the boss of this period. Everything else works around it.”

Procedure

Step	Teacher will	Students will	Time
Prep phase (0 to 10)	Start the timer when every station is standing. Walk the	Station Manager checks the drawer map and sets out 7	10 min

	room once checking hair, sleeves, hands, and surfaces. Stay at the stove row. Say nothing about the recipe unless asked; the station card and the role cards are the teacher now. Confirm every pot is on the burner and on high by minute 3, because that is the critical path and three minutes late here is three minutes late at the end.	plates, 7 forks, napkins. Head Cook and Helper Cook fill the pot to three quarters, carry it with two people, set it on the burner handle in, turn it to HIGH, and add 1 Tbsp of salt. Second Cook opens the marinara, pours it into the saucepan, and sets the burner to LOW at minute 8. Prep Cook measures 1 1/2 cups of cheese into a bowl and levels it. Toppings Chief portions a small cup of cheese per plate. Plate Builder sets the serving bowl and the tongs and puts the colander in the sink. Everyone: mise en place complete before any pasta goes in.	
Cook phase (10 to 28)	Stand at the stove row the whole time. Watch for a pot handle turned out, a towel near a burner, and sauce on high. Call time at 15, 20, and 25. At the drain, clear the path to the sink, take the pot yourself with both mitts, pour it into the colander, and shake it twice. Say what you are doing	Head Cook watches for a rolling boil (big bubbles breaking the surface, not small ones at the bottom), adds the pasta at the boil, and stirs immediately, then every two to three minutes. Time Keeper or Station Manager sets the box time minus one minute and calls it. Second Cook stirs	18 min

	while you do it.	the sauce every two minutes and keeps it on LOW, never boiling. Helper Cook tastes one piece at the timer with a clean fork and calls soft with a little bite or not yet. At the call, the Head Cook turns the burner off and steps back; the teacher drains. Head Cook returns the drained pasta to the pot or the serving bowl, Second Cook pours the sauce over and tosses gently. Station Manager washes prep dishes as they free up.	
Plate and eat (28 to 35)	Walk the room. Ask one question per station from the “deepen” list while they eat. Score the Lab Rubric on the clipboard as you go: Objectives 1 and 2 and the safety row.	Plate Builder serves seven portions with the tongs. Toppings Chief sprinkles cheese on each plate while the pasta is still hot so it melts. Everybody eats. The Station Manager counts plates before anyone sits, so no teammate is missed.	7 min
Clean (35 to 40)	Call “clean” at 35. Check each station against the clean-up zone chart. Sign off only when every line is done. Nobody leaves until every station is signed.	Each pair works its clean-up zone (stove and pots; counters and supplies; dishes and sink; floor and trash). Station Manager walks the station check: burners off, pot and saucepan and	5 min

		colander washed and dry and in the drawer per the map, stovetop wiped, counters wiped, sink empty, floor swept, trash out, fridge closed.	
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Questions to ask

Monitor understanding:

- Is that a rolling boil or is it almost there? What are you looking at?
- Your sauce is on medium. What happens to tomato sauce on medium for ten minutes?
- Whose hands are allowed on that pot right now?
- What is your role's next job, and what time does it start?

Deepen learning:

- The water took eight minutes and you could not do anything about it. What did your station do with those eight minutes?
- This whole meal fed seven people for about ten dollars. What is that per person, and how does it compare to a slice of pizza and a drink?
- The pasta is a dry heat food or a moist heat food? Prove it with what happened in the pot.
- If I told you the period was thirty minutes instead of forty, what would you cut and what would you start earlier?

Check for understanding

The clipboard Lab Rubric during plate, eat, and clean, plus the “what does done look like” checklist printed on the station card, which the station checks itself at the end of each phase. One verbal check per station during the cook phase: “name the critical path task.” A “got it” answer is the boiling and draining of the pasta, with a reason about what waits on it.

Closure (Time: 2 min, inside the clean phase)

The station check is the closure. When I sign a station, the Station Manager reads the team's answer to one question out loud: “did we keep the flowchart, and where did we lose or gain time?” The Lab Reflection is completed in the first five minutes of Lesson 6.9 or goes home.

Differentiation and supports

- ELL: the station card has a picture beside each step and every abbreviation spelled out; role cards are short and organized by phase so they can be scanned rather than read; the phrases “behind you,” “hot,” “boiling,” “done,” and “help” are on the station

card in Spanish (see Vocabulary 06.md); pair each ELL student with a role that has a tag-in partner (Helper Cook with Head Cook).

- IEP and 504 (general): a student who should avoid heat takes Prep Cook, Toppings Chief, or Plate Builder, none of which touch the stove; a large visible timer; the role card is a checklist to tick; a student who cannot stand for forty minutes works the Toppings or Plate role from a stool at the counter; sensory note: the pot lid rattles and the drain hisses, and both get announced before they happen.
- Allergy substitutions: wheat, one box of gluten free pasta cooked in a separate small pot with its own colander, drained by me first so no cross contact; dairy, a dairy free shred for that portion and a plate set aside before any cheese goes near it; soy, check the marinara label the day you shop and pick a jar without it. Any student with an allergy takes a role that does not handle the allergen.
- Grade 6 support: elbow macaroni, not spaghetti, because it does not need to be broken or twirled and it drains faster. The station card for grade 6 lists one step per line with a picture and no combined steps.
- Grade 8 stretch: the Time Keeper role runs as a scheduler, drawing the station's actual times on a blank flowchart during the lab and reporting the variance at the station check. Stretch students also compute the cost per serving from the receipt and answer what one dollar more per station would best be spent on.
- UDL checkpoint used: multiple means of action and expression (the role card can be followed by reading, by picture, or by the tag-in partner's example) and multiple means of engagement (the flowchart is a visible clock the station competes against, and the station check is a pass or not-yet that the team controls).
- No-kitchen alternative: Cold Noodle Bowl, assembled at the station, run with the same eight roles, the same four phases, and the same station check. Per station of seven: 1 lb pasta cooked the day before and chilled (or a 12 oz bag of precooked or shelf stable noodles), 1 cup shredded mozzarella or cubed cheese, 1 can of drained corn, 1 can of drained black beans, 1 bottle of Italian dressing or a jar of pesto, 7 bowls and forks. Prep 0 to 10 is the same. "Cook" 10 to 28 becomes measure, combine, toss, and portion, with the Head Cook running the toss and the Second Cook running the dressing measurement. Plate 28 to 35 and clean 35 to 40 are unchanged. A second option is a no-cook pasta salad with the same ingredients and a vinegar and oil dressing measured by the Prep Cook, which keeps the measuring objective intact. Cost about \$11 per station.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. State the role's tasks by phase and name the critical path	FSP 5. d); CIR 3. f); HSW 4. c)	Role card, do now, and the verbal check in the cook phase	Formative; verbal check at the station, one question per station; the do now answer collected

2. Prepare and serve the pasta safely and on schedule	FSP 1. a), c), e); Std 1 PI A and PI B; National FCS 14.3.3, 8.5.7, 14.4.2; CDOS 3a Managing Resources	The lab, all four phases	Summative; Rubric 01 - Lab Rubric.md, three objective rows plus the safety and sanitation row, 16 points, scored on the clipboard during plate, eat, and clean
3. Judge the station against the flowchart and recommend one change	FSP 5. d); CIR 3. f); HSW 4. e)	Station check answer and the Lab Reflection	Formative; Handout 01.08 - Lab Reflection.md scored complete, partial, or missing; the recommendation must name a role and a minute. Feeds Lesson 6.9 and the Feed the Class project flowchart in Lesson 6.14

Homework

The Lab Reflection, if it is not finished in class. Optional: cook pasta at home for somebody and time how long the water took to boil on your stove.

Connections

Math: cost per serving from the receipt, and the flowchart is a real scheduling problem with a critical path, which is the same idea a project management unit teaches. Science: the pasta absorbs water and swells, which is why a pound of dry pasta feeds seven, and Lesson 6.6's moist heat lesson explains why it never browns. Careers: a line cook works exactly this way, and the critical path language is what a kitchen manager uses.

Sources

- NYS Middle Level CTE FACS Content Module 06, Food Systems and Production (nyctecenter.org, June 2018), lines 1. a), 1. c), 1. e), and 5. d). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 04, Health, Safety, and Wellness, lines 4. c) and 4. e); Theme Module 02, Communication and Interpersonal Relationships, line 3. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 1, Intermediate, FACS strand, Performance Indicators A and

B. Wording per 02 Standards/STANDARDS SOURCES - web research.md, section 2.

- CDOS Intermediate Standard 3a Universal Foundation Skills, Managing Resources. Same file, section 3.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 14.3.3, 8.5.7, and 14.4.2.
- Rubric 01 - Lab Rubric.md and Handout 01.08 - Lab Reflection.md, reused unchanged.
- My own materials: My package supplied the four-phase workflow, the Gantt chart by role, the critical path analysis, the eight role cards with tag-in notes, the clean-up zone assignments, the supermarket shopping list with budget swaps, the “what could go wrong” boxes, the teacher facilitation notes, and the reflection worksheet. Rebuilt here for a station of seven, with the standards table replaced by correct citations and the no-bake pudding dessert held for the no-kitchen option and for the Feed the Class project.

Teacher notes

- The pot goes on the burner by minute 3 or the period does not work. That is the only sentence you need to repeat during prep.
- Fill to three quarters. A full pot is heavy, slow, and it boils over. Two students carry it, not one.
- I drain. Every time, all year. Say it while you do it so the room hears the rule and the reason: steam burns deeper than water.
- My own “what could go wrong” notes, kept because they are right: pasta stuck in a clump means it was not stirred right after it went in, so stir immediately and again every two to three minutes; mushy pasta means it was overcooked, so set the timer for one minute less than the box and taste; sauce on anything above low will scorch on the bottom and taste burnt through the whole pan.
- Common mistake: the Head Cook adds pasta to water that is not boiling yet. Small bubbles at the bottom are not a boil. Show the class one rolling boil at the front before the lab if you have a spare minute in Lesson 6.7.
- My package included a no-bake chocolate pudding dessert running in parallel. It is dropped from this lab on purpose, to keep one dish and one clock and to keep the critical path visible, which is the point of the day. The pudding fits as a dessert station in the Feed the Class project (Lesson 6.16) or as the grade 8 second dish in the no-kitchen version.
- The Time Keeper role is written for eight students. At seven, the Station Manager keeps time, and the card says so.
- Per-station cost estimate (store brand, Long Island supermarket; every price to be refreshed from the current circular before teaching):

Item	Qty per station of seven	Estimated price	Budget swap
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Pasta, 1 lb, store brand	1 box	\$1.29	All store brands are about the same
Marinara sauce, 24 oz, store brand	1 jar	\$2.49	Store brand tomato sauce plus Italian seasoning, about \$1.29
Shredded mozzarella, 8 oz	2 bags	\$2.99 each, \$5.98	1 1/2 bags of store brand mozzarella, about \$3.74
Salt	1 Tbsp	pantry	
Estimated total per station		about \$9.76	about \$6.32 with swaps
Class of four stations		about \$39	about \$25 with swaps. Most unprepared grocery food is exempt from New York State sales tax.
No-kitchen Cold Noodle Bowl	per station of seven	about \$11	Pasta \$1.29, cheese \$2.99, corn \$0.99, black beans \$0.99, dressing \$2.49, bowls and forks about \$2

- Both versions are well under the \$20 per station cap. The unit budget line for Lab 3 should carry about \$40 for a class of four stations at full prices.